جامعة ابن طفيل
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TP 5: Apache Spark

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Installation de Apache Spark

Étape 1 : téléchargement

1. Installing Java(JDK)

- java is already installed

```
❯ java -version
openjdk version "1.8.0_482"
OpenJDK Runtime Environment (build 1.8.0_482-8u482-ga~us1-0ubuntu1~24.04-b08)
OpenJDK 64-Bit Server VM (build 25.482-b08, mixed mode)
Spark ❯
```

2. installing Apache Spark

```
❯ wget https://downloads.apache.org/spark/spark-4.1.1/spark-4.1.1-bin-hadoop3.tgz

--2026-04-19 14:25:16--  https://downloads.apache.org/spark/spark-4.1.1/spark-4.1.1-bin-hadoop3.tgz
Resolving downloads.apache.org (downloads.apache.org)... 88.99.208.237, 135.181.214.104, 2a01:4f9:3a:2c57::2, ...
Connecting to downloads.apache.org (downloads.apache.org)|88.99.208.237|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 572739279 (546M) [application/x-gzip]
Saving to: ‘spark-4.1.1-bin-hadoop3.tgz’

spark-4.1.1-bin-had 100%[===================>] 546.21M  1.21MB/s    in 15m 53s

2026-04-19 14:41:10 (587 KB/s) - ‘spark-4.1.1-bin-hadoop3.tgz’ saved [572739279/572739279]

❯ ls
installation  spark-4.1.1-bin-hadoop3.tgz
Spark ❯
```

Installation de Apache Spark

Étape 2: Configuration des Variables d'Environnement

1. Extract the file

```
mohammmed@ROG-Strix:~/Spark
❯ tar -xvf spark-4.1.1-bin-hadoop3.tgz
spark-4.1.1-bin-hadoop3/
spark-4.1.1-bin-hadoop3/conf/
spark-4.1.1-bin-hadoop3/conf/workers.template
spark-4.1.1-bin-hadoop3/conf/fairscheduler.xml.template
spark-4.1.1-bin-hadoop3/conf/spark-defaults.conf.template
spark-4.1.1-bin-hadoop3/conf/log4j2.properties.template
spark-4.1.1-bin-hadoop3/conf/log4j2-json-layout.properties.template
spark-4.1.1-bin-hadoop3/conf/spark-env.sh.template
spark-4.1.1-bin-hadoop3/conf/metrics.properties.template
spark-4.1.1-bin-hadoop3/NOTICE
spark-4.1.1-bin-hadoop3/jars/
spark-4.1.1-bin-hadoop3/jars/datasketches-memory-3.0.2.jar
spark-4.1.1-bin-hadoop3/jars/jline-2.14.6.jar
spark-4.1.1-bin-hadoop3/jars/kubernetes-model-autoscaling-7.4.0.jar
spark-4.1.1-bin-hadoop3/jars/javassist-3.30.2-GA.jar
spark-4.1.1-bin-hadoop3/jars/spark-network-shuffle_2.13-4.1.1.jar
spark-4.1.1-bin-hadoop3/jars/netty-handler-4.2.7.Final.jar
spark-4.1.1-bin-hadoop3/jars/orc-shims-2.2.1.jar
spark-4.1.1-bin-hadoop3/jars/hadoop-client-api-3.4.2.jar
spark-4.1.1-bin-hadoop3/jars/spark-yarn_2.13-4.1.1.jar
spark-4.1.1-bin-hadoop3/jars/avro-1.12.1.jar
spark-4.1.1-bin-hadoop3/jars/spark-kvstore_2.13-4.1.1.jar
spark-4.1.1-bin-hadoop3/jars/jakarta.servlet-api-5.0.0.jar
```

2. Move Spark to `~/`

```
mohammmed@ROG-Strix:~/Spark
❯ mv spark-4.1.1-bin-hadoop3 ~/spark
Spark ❯
```

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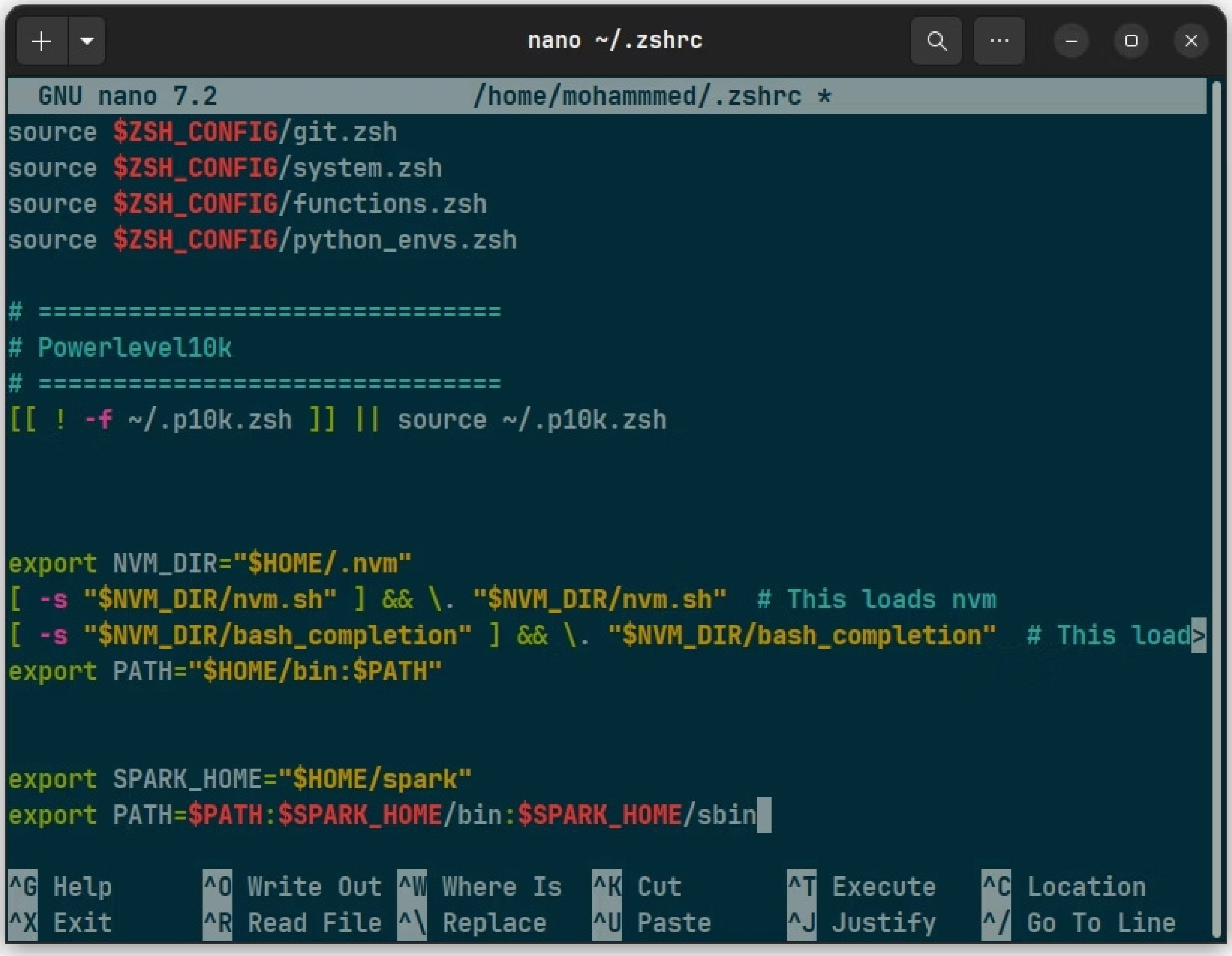
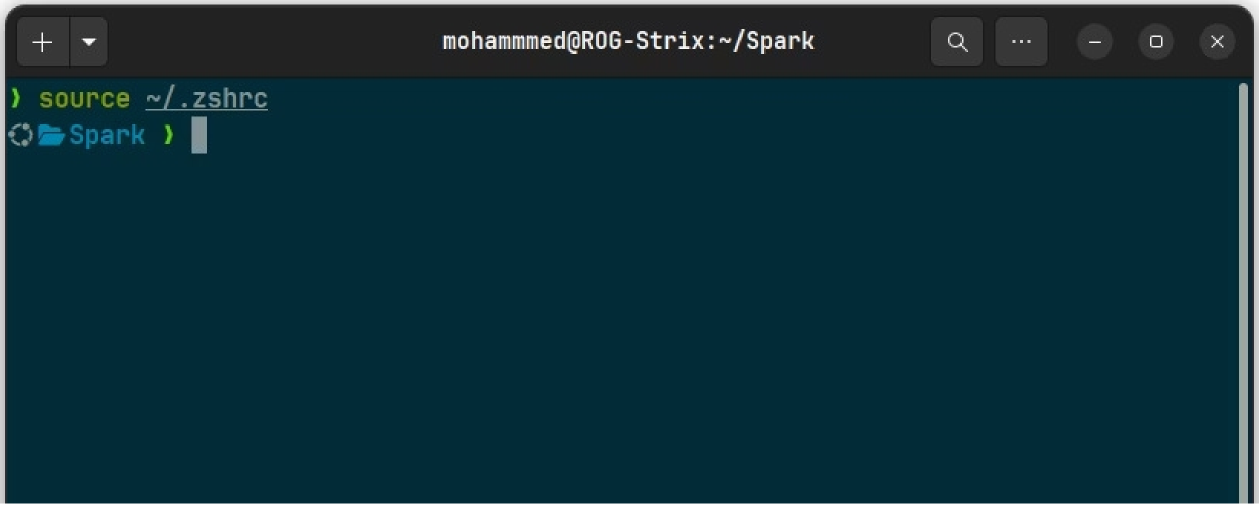
Étape 3: Configuration des Variables d'Environnement

3. Set environment variables



3. Apply changes



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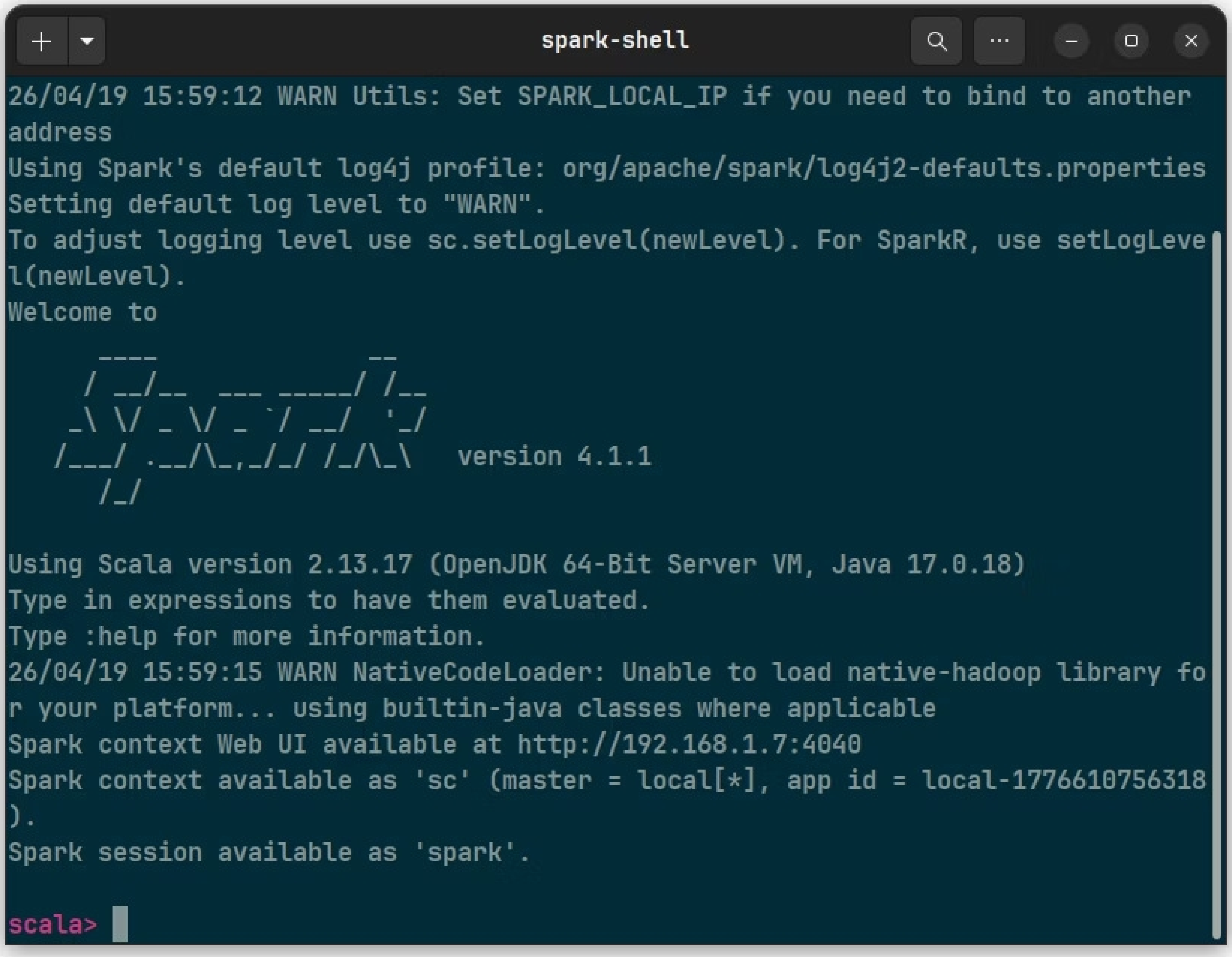
Étape 2: Vérification et Lancement



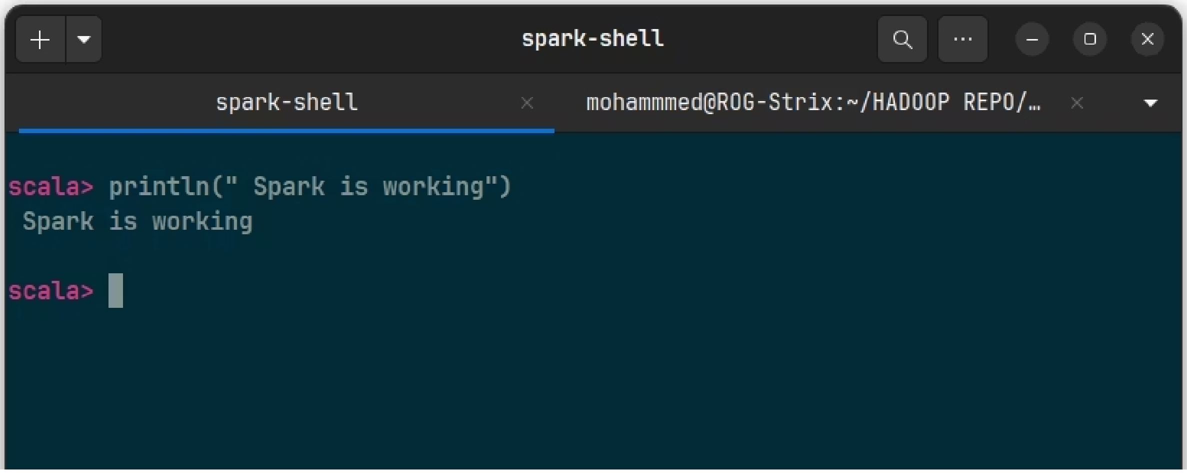
```
spark-shell
26/04/19 15:59:12 WARN Utils: Set SPARK_LOCAL_IP if you need to bind to another
address
Using Spark's default log4j profile: org/apache/spark/log4j2-defaults.properties
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLeve
l(newLevel).
Welcome to
      ____              __
     / __/__  ___ _____/ /__
    _\ \/ _ \/ _ `/ __/  '_/
   /___/ .__/\_,_/_/ /_/\_\   version 4.1.1
      /_/

Using Scala version 2.13.17 (OpenJDK 64-Bit Server VM, Java 17.0.18)
Type in expressions to have them evaluated.
Type :help for more information.
26/04/19 15:59:15 WARN NativeCodeLoader: Unable to load native-hadoop library fo
r your platform... using builtin-java classes where applicable
Spark context Web UI available at http://192.168.1.7:4040
Spark context available as 'sc' (master = local[*], app id = local-1776610756318
).
Spark session available as 'spark'.

scala>
```

- Vérification :

```
Println("Spark is working")
```



```
spark-shell
spark-shell    mohammmed@ROG-Strix:~/HADOOP REPO/…

scala> println(" Spark is working")
 Spark is working

scala>
```

Manipulation des packages

1. filtrer ceux qui contiennent la lettre "a"

```
spark-shell
spark-shell    mohammmed@ROG-Strix:~/HADOOP REPO/…

scala> val data = sc.parallelize(List("Hadoop", "Spark", "BigData", "Zineb", "Im
ane"))
data: org.apache.spark.rdd.RDD[String] = ParallelCollectionRDD[0] at parallelize
 at <console>:23

scala> val result = data.filter(name => name.contains("a")).collect()
result: Array[String] = Array(Hadoop, Spark, BigData, Imane)

scala>

scala> result
res0: Array[String] = Array(Hadoop, Spark, BigData, Imane)

scala>
```

2. Trouver les mots qui commencent par "S"

```
spark-shell
spark-shell    mohammmed@ROG-Strix:~/HADOOP REPO/…

scala> val result = data.filter(name => name.startsWith("S")).collect()
result: Array[String] = Array(Spark)

scala>
```